

ASSIGNMENT

2. Take one Domain and draw the graph (Normal distribution) (Empirical rule).

Introduction to Normal Distribution:

- The normal distribution is a common continuous probability distribution in statistics.
- It is characterized by its bell-shaped curve that is symmetric around the mean.
- Understanding this distribution is essential for analyzing many real-world phenomena.

Importance of the Normal Distribution:

- Many natural and social phenomena follow a normal distribution pattern.
- It helps in making predictions and decisions based on data.
- This distribution forms the foundation for many statistical inference techniques.

STATS 113 Problem Sessions

Normal Distribution/Empirical Rule/Z-Scores Word Problems

1. The average playing time of CDs in a large collection is 35 minutes, and the standard deviation is 5 minutes.
 - a. What value is 1 standard deviation above the mean? 1 standard deviation below the mean? What values are 2 standard deviations from the mean?
 - b. Assuming the distribution of time is approximately normal, about what percentage of times are between 25 and 45 minutes? Less than 20 minutes or greater than 50 minutes? Less than 20 minutes?
 - a) 40 minutes, 30 minutes, 25-45 minutes
 - b) 95%, 0.3%, 0.15%
2. Suppose your statistics professor returned your first midterm exam with only a z-score written on it. She also told you the histogram of the scores was approximately normal. How would you interpret each of the following z-scores?
 - a. 2.2
 - b. -0.4
 - c. 0
 - a) You did very well!
 - b) You did slightly below average.
 - c) You did average.
3. The test scores for the quantitative reasoning section of the GRE are normally distributed. The mean score is 150 with a standard deviation of 8.75. The test scores of four students are 162, 168, 155, and 138. Find the z-score that corresponds to each value.

Score 162 $\rightarrow z = 1.37$	Score 155 $\rightarrow z = 0.57$
Score 168 $\rightarrow z = 2.06$	Score 138 $\rightarrow z = -1.37$
4. In a recent study on world happiness, participants were asked to evaluate their current lives on a scale from 0 to 10, where 0 represents the worst possible life and 10 represents the best possible life. The responses were normally distributed, with a mean of 5.4 and a standard deviation of 2.2. Find the probability that a randomly selected study participant's response was:
 - a. Less than 4
 - b. Between 4 and 6
 - c. More than 8
 - a) 0.2611
 - b) 0.3453
 - c) 0.1190

Key Features of a Normal Distribution:

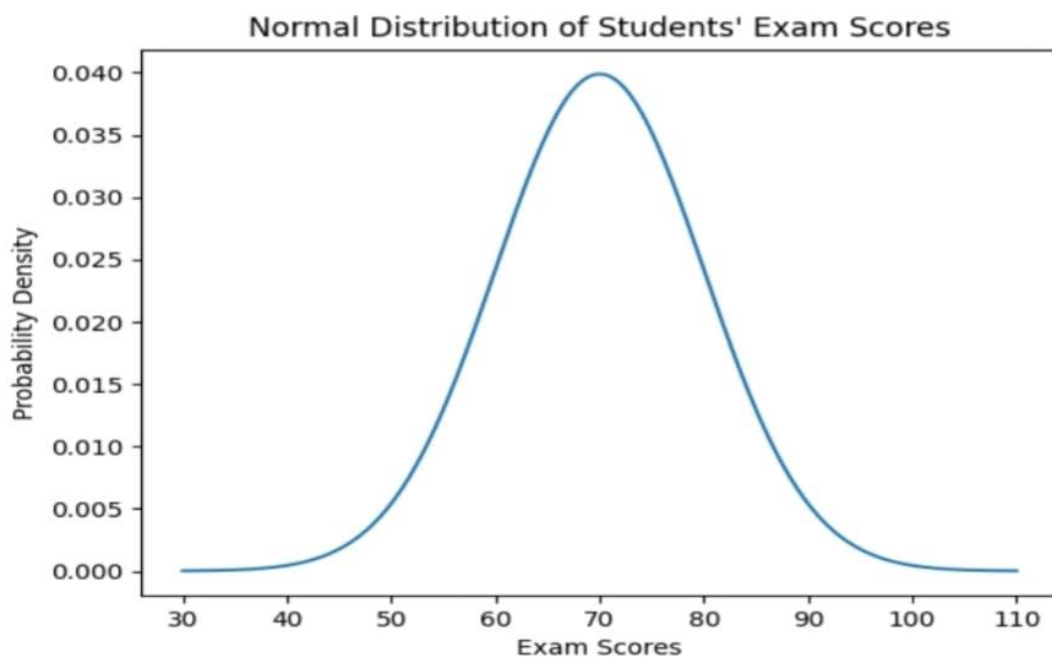
- The curve is symmetric about the mean, median, and mode.
- The total area under the curve equals 1, representing probability.
- The shape is determined by the mean (μ) and standard deviation (σ).

Domain Chosen: Students' Exam Scores

In this report, we assume that students' exam scores follow a normal distribution with a mean (μ) of 70 and a standard deviation (σ) of 10.

Empirical Rule (68-95-99.7 Rule):

- 68% of students score between 60 and 80 ($\mu \pm 1\sigma$)
 - 95% of students score between 50 and 90 ($\mu \pm 2\sigma$)
 - 99.7% of students score between 40 and 100 ($\mu \pm 3\sigma$)
- The graph below represents the normal distribution curve for the selected domain.



- Normal distribution is one of the most important probability distributions in statistics.
- It is symmetric, bell-shaped, and defined by its mean and standard deviation.
- Many real-world variables such as exam scores, heights, and measurement errors follow this distribution.

Characteristics of Normal Distribution:

- Symmetrical about the mean
- Mean = Median = Mode
- Total area under the curve equals 1
- Defined by mean (μ) and standard deviation (σ)
- Bell-shaped curve.

Mathematical Representation:

The probability density function of normal distribution is given by:

$$f(x) = (1 / (\sigma\sqrt{2\pi})) e^{-(x-\mu)^2 / 2\sigma^2}$$

Where μ is the mean and σ is the standard deviation.

Empirical Rule (68-95-99.7 Rule):

The Empirical Rule describes how data is distributed in a normal distribution:

- 68% of data lies within $\mu \pm 1\sigma$
- 95% of data lies within $\mu \pm 2\sigma$
- 99.7% of data lies within $\mu \pm 3\sigma$

Application to Students' Exam Scores:

Assume Mean = 70 and Standard Deviation = 10

- 68% of students score between 60 and 80.
- 95% of students score between 50 and 90.
- 99.7% of students score between 40 and 100.

Importance in Education:

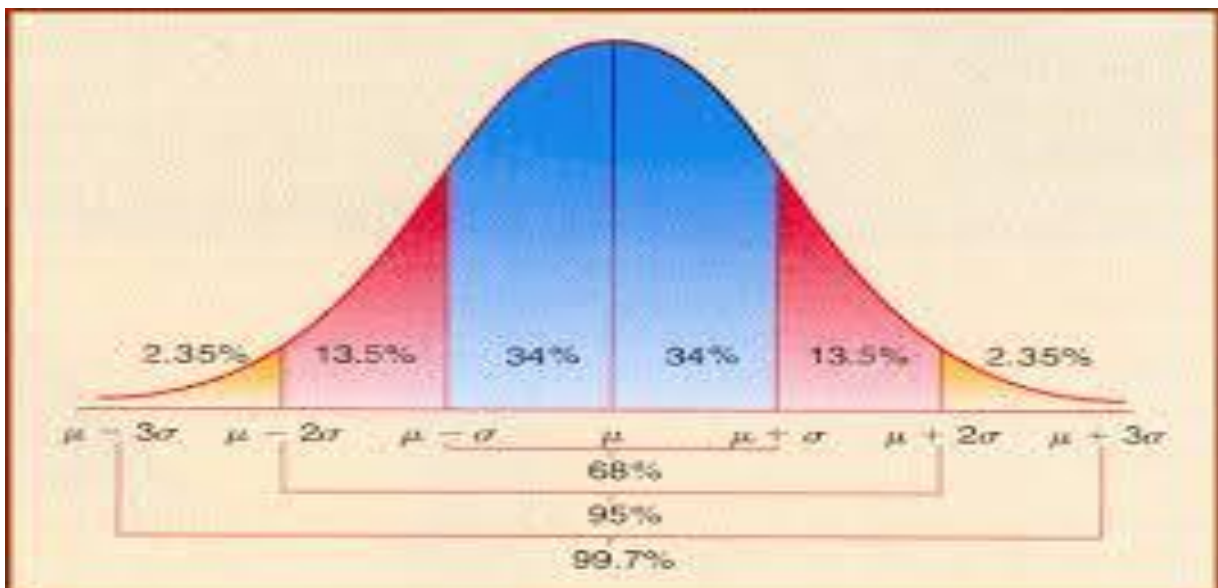
- Normal distribution helps teachers analyze performance trends, identify outliers, and standardize grading systems.
- It ensures fairness and consistency in evaluation.

Advantages:

- Easy to analyze and interpret.
- Widely applicable in research.
- Forms basis of many statistical tests.
- Predictable probability intervals.

Defining the Domain:

- The domain of a normal distribution is all real numbers from negative to positive infinity.
- The domain covers every possible value the random variable can take.
- In practice, most data falls within a few standard deviations from the mean.



Mathematical Equation of the Normal Distribution:

- The probability density function (PDF) is given by a specific formula involving μ and σ .

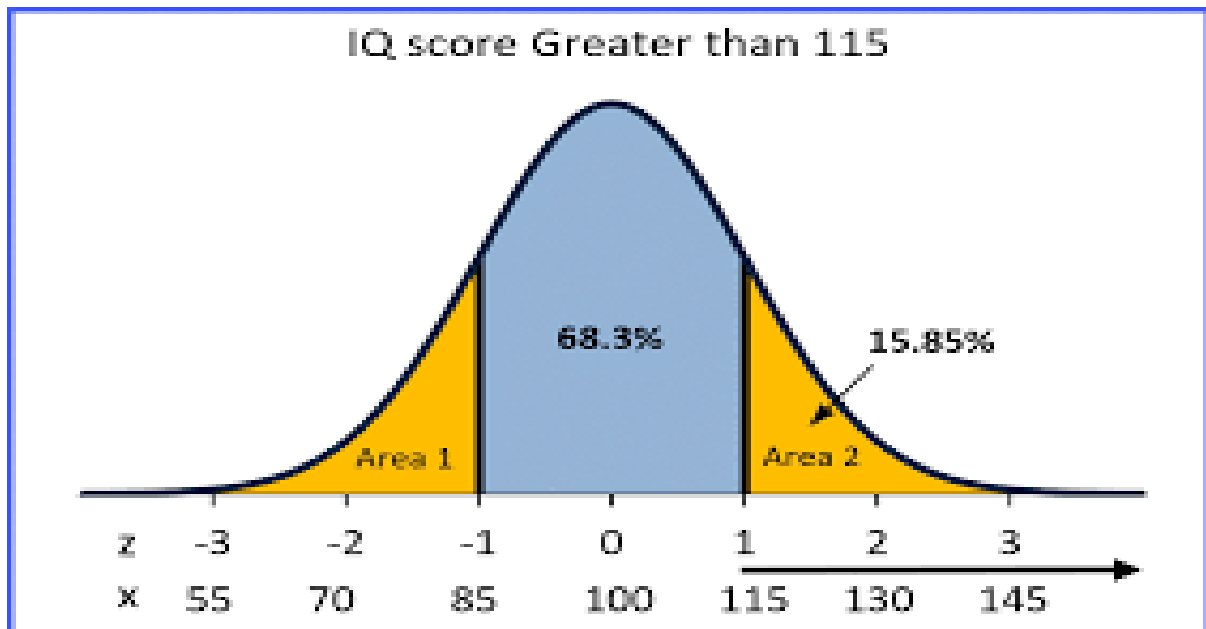
- The formula includes an exponential function to describe the bell shape.
- Understanding this formula helps in plotting the distribution accurately.

Drawing the Normal Distribution:

- Graph To draw the graph, identify the mean and standard deviation of your data.
- Plot the curve symmetrically around the mean, with height determined by the PDF.
- Use computer software or graphing tools for precise and accurate visualization.

Example Domain: Heights of Adults:

- Consider the heights of adult males in a population.
- Assume a mean height of 70 inches with a standard deviation of 3 inches.
- This data typically follows a normal distribution pattern.



Plotting the Example Data:

- Mark the mean (70 inches) at the center of your graph.

- Draw the bell-shaped curve, tapering off as you move away from the mean.
- Highlight the areas within 1, 2, and 3 standard deviations.

Empirical Rule Introduction:

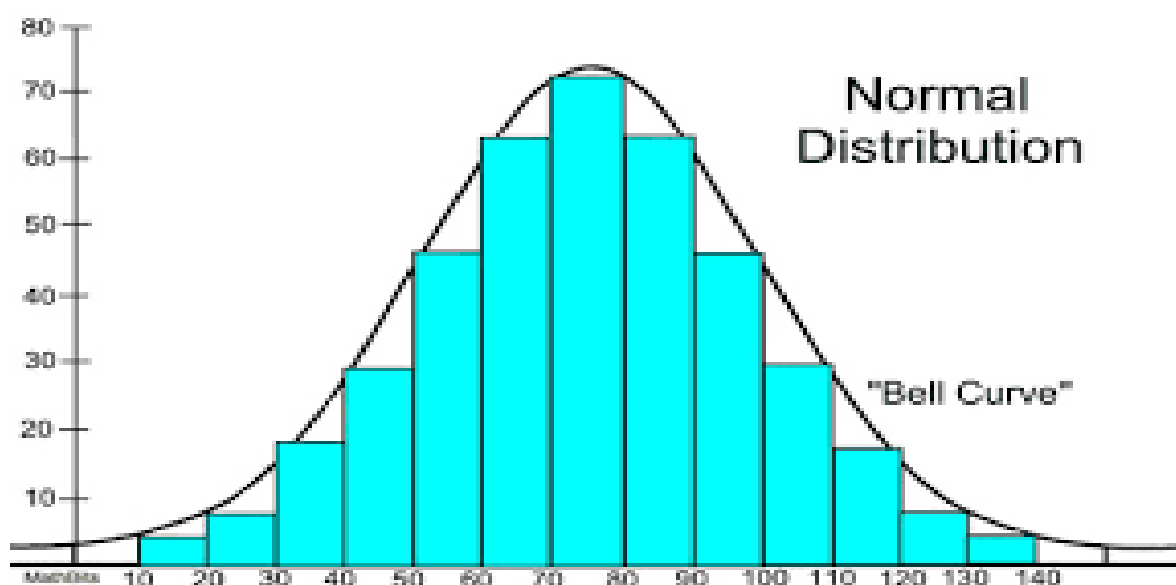
- The empirical rule provides approximate percentages of data within certain standard deviations.
- It applies specifically to normal distributions.
- It simplifies understanding how data is spread around the mean.

The Empirical Rule:

- Breakdown About 68% of data falls within 1 standard deviation of the mean.
- Approximately 95% lies within 2 standard deviations.
- Nearly 99.7% of data is within 3 standard deviations.

Visualizing the Empirical Rule:

- Shade the regions under the curve corresponding to 1, 2, and 3 standard deviations.
- These shaded areas visually demonstrate the percentage of data within each region.
- It helps in quickly assessing data distribution in real-world scenarios.



Applying the Empirical Rule to Heights:

- For the height example, 68% of males are between 67 and 73 inches.
- 95% are between 64 and 76 inches.
- Nearly all (99.7%) are between 61 and 79 inches.

Advantages of Using the Empirical Rule:

- It provides quick estimates without detailed calculations.
- Useful for identifying outliers outside three standard deviations.
- Helps in assessing the normality of data visually and statistically.

Checking Normality in Data:

- Use histograms and normal probability plots to assess data distribution.
- If the data aligns well, the empirical rule applies.
- Deviations suggest the distribution may not be normal.

Limitations of the Empirical Rule:

- It is only an approximation and assumes a normal distribution.
- Not applicable for skewed or bimodal data.
- Always verify normality before applying the empirical rule in analysis.

Practical Applications of Normal Distribution:

- Used in quality control and process management.
- Applied in finance for stock return analysis.
- Important in psychology, biology, and social sciences research.

Tools for Drawing Normal Distributions:

- Software like Excel, R, and Python can generate accurate plots.
- Graphing calculators can also produce normal distribution curves.
- Using digital tools ensures precision and ease of visualization.

Summary of Key Concepts:

- The normal distribution is symmetric and defined by its mean and standard deviation.
- The empirical rule provides quick estimates of data spread.

- Accurate visualization helps in understanding data distribution and variability.

Recap: Takeaways:

- Drawing the normal distribution involves understanding its shape and parameters.
- The empirical rule is a handy tool for interpreting data spread in normal distributions.
- Combining visualization and statistical rules enhances data analysis skills.

Limitations:

- Not all data is normally distributed.
- Sensitive to extreme values.
- Assumes symmetry which may not exist in real data.

Conclusion:

- Normal distribution and the Empirical Rule are fundamental concepts in statistics.
- Using students' exam scores as a domain, we demonstrated how data clusters around the mean and how probability intervals help in understanding performance distribution.